

Physical Quantities and Vectors Revision Questions

- a) Convert 52 miles per hour to meters per second. Given that 1 mile = 1609 m.
- b) The forces shown in Figure 1 are in equilibrium. Find the resultant force and its direction.

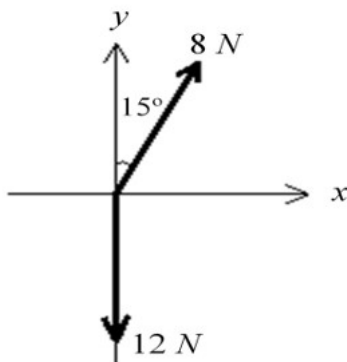


Figure 1

- c) Given vector F_1 and F_2 are in the x-y plane. F_1 is 60 N at 48° and F_2 is 90 N at 110° counterclockwise. Find the magnitude and direction of the resultant vector. Ans: 129.5 N, 85.9° Q1.
- d) Two forces of magnitudes 3.0 N and 4.0 N act on an object. Determine how the directions (same directions, opposite direction or perpendicular direction) of the two forces related if
- the net force has magnitude 7.0 N,
 - the net force has magnitude 5.0 N.
 - Determine the relationship between the directions gives the smallest magnitude of net force and calculate the magnitude.
- e) Convert 1.0 kg/m^3 to unit of $\mu\text{g/cm}^3$.
- f) Change the following value $5 \mu\text{m}^3/\text{hour}$ to unit m^3/s .
- g) The area of a land is 500 km^2 . Express this area in mi^2 . Given $1 \text{ mi} = 1.6 \text{ km}$.
- h) Figure B below shows the forces exerting on an object.

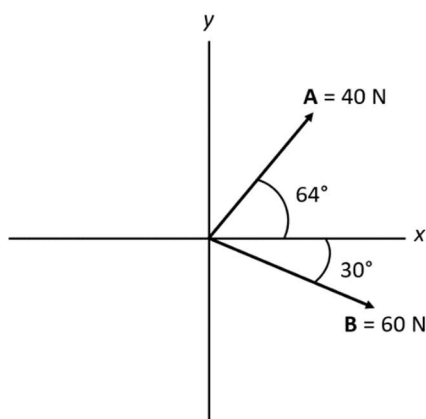


Figure B

- i) Find the x -component and y -component for vector **A**.
 - ii) Find the x -component and y -component for vector **B**.
 - iii) Calculate the net value for the x -components.
 - iv) Calculate the net value for the y -components.
 - v) Calculate the magnitude of the resultant vector.
 - vi) Determine the direction of the resultant vector.
- i) A certain fuel-efficient hybrid car has gasoline mileage of 65 mpg (miles per gallon). Given 1 gallon = 3.788 liters, 1 miles = 1.609 km.
- (i) If you are driving this car in Europe and want to compare its mileage with that of other European cars, express this mileage in km/L (L = liter).
 - (ii) If this car's gas tank holds 40 L, calculate the number of tanks of gas you will use to drive 1800 km.
- j) Find the net force acting on the system shown in Figure 2. Given forces F_1 , F_2 , F_3 and F_4 are 5 N each.

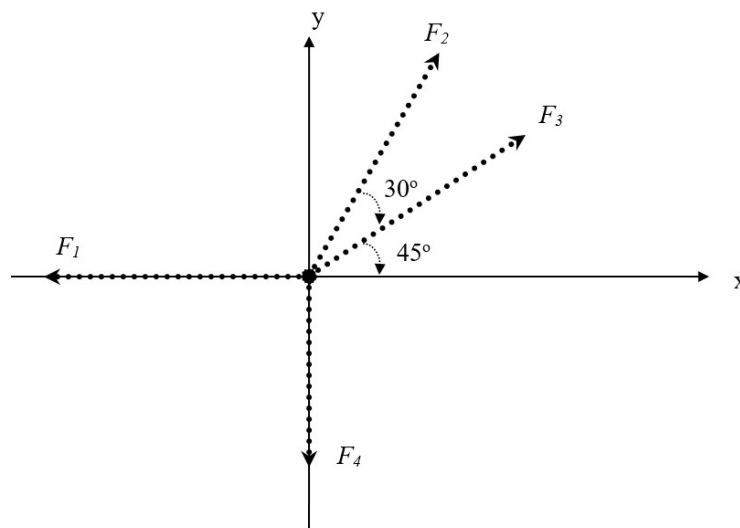
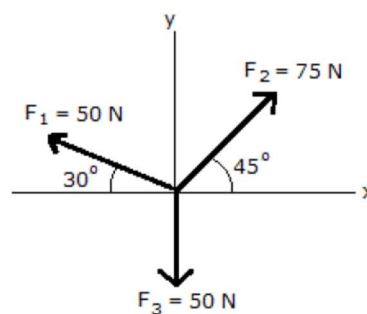


Figure 2

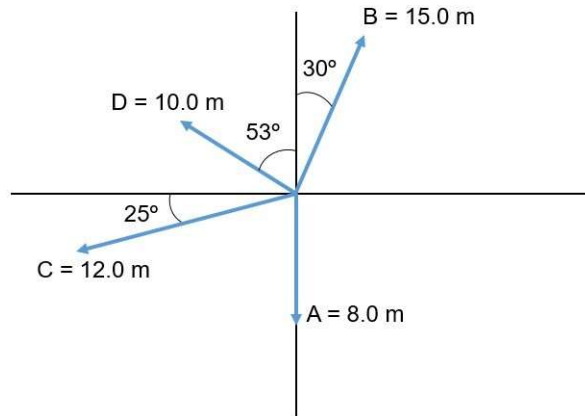
- k) Determine the magnitude and direction of the resultant force by adding the rectangular components of the three forces in Figure below.



l) Convert the following to SI units:

- (i) 76.5 cm / ms
- (ii) 150 ng / mm³

m) Figure below shows vectors **A**, **B**, **C** and **D** with their respective directions.



- (i) Determine the x and y components for each vectors.
- (ii) Find the magnitude and direction of the resultant vector.

n) Consider the vectors **A**, **B** and **C** shown in **Figure 5** below.

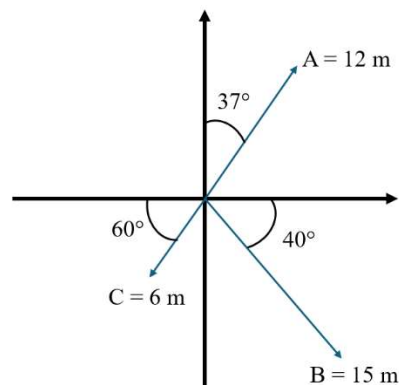


Figure 5

- i) Find the x -component and y -component for vector **A**.
 - ii) Find the x -component and y -component for vector **B**.
 - iii) Find the x -component and y -component for vector **C**.
 - iv) Calculate the net value for the x -components.
 - v) Calculate the net value for the y -components.
 - vi) Calculate the magnitude of the resultant vector.
 - vii) Determine the direction of the resultant vector.
- o) What is the resultant force on the box in **Figure 1** below?



Figure 1

- p) Convert 150 g/cm^3 to kg/m^3 .
- q) Mr. Bean is considering buying a European car and wants to see if its advertised fuel efficiency (expressed in km/L) is better than that of his present car. If his car gets 37.3 miles per gallon, how many km/L is this? Given 1 gallon = 3.788 liters, 1 miles = 1.609 km.
- r) Position as a function of time equation is given as:

$$x(t) = 2.5t^3 - 6.1t^2 - 3t + 9$$
Determine the instantaneous acceleration of the motion at $t = 1.5 \text{ s}$.

Answers:

- a) 23.24 m/s
- b) 4.75 N, 64° Q4
- c) 129.5 N, 85.9° Q1.
- d) i) Same directions ii) Perpendicular directions iii) Opposite directions, 1 N
- e) $1000 \mu\text{g/cm}^3$
- f) $1.39 \times 10^{-21} \text{ m}^3/\text{s}$
- g) 195.3 mi²
- h) i) +17.53 N, +35.95 N ii) +51.96 N, -30 N iii) 147.53 N
iv) 21.96 N v) 149.16 N vi) 8.47° at the Q1
- i) i) 27.61 km/L ii) 2 tanks
- j) 3.37 N, 87.11° at Q2
- k) 24.03 N, 66.1° at Q4
- l) i) 765 m/s ii) 0.15 kg/m^3
- m) i) $A_x = 0$ ii) 12.83 m, 27.58° at Q2
 $A_y = -8\text{m}$
 $B_x = 7.5\text{m}$
 $B_y = 12.99\text{m}$
 $C_x = -10.88\text{m}$
 $C_y = -5.07\text{m}$
 $D_x = -7.99\text{m}$
 $D_y = 6.02\text{m}$
- n) i) $A_x = 7.22 \text{ m}$, $A_y = 9.58 \text{ m}$
ii) $B_x = 11.49\text{m}$, $B_y = -9.64\text{m}$
iii) $C_x = -3\text{m}$, $C_y = -5.2\text{m}$
iv) 15.71m
v) -5.26m
vi) 16.57m
vii) 18.51° at Q2
- o) 1N to the left
- p) $1.5 \times 10^5 \text{ kg/m}^3$
- q) 15.8 km/L
- r) 10.3 m/s^2